Supplementary Material

Hierarchical RF Tokenization for Parameter-Efficient Cross-Receiver LoRa Identification

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13 November 2026

This supplement records the protocol, controls, data roles, numerical qualifications, and inherited components of the consolidated study. It introduces no new training, model selection, or receiver scoring. The numerical CSV files are distributed with the source bundle and retain their original precision.

1 Receiver Protocol and Exposure Ledger

The frozen protocol is P-H5x160. The corpus has ten transmitters with original labels 31–40, twenty SDR receivers, 1 MHz sample rate, nominal 125 kHz LoRa bandwidth, spreading factor 7, and 8,192-complex-sample windows. Models do not consume the dataset's CFO or SNR fields.

Fold	Source RX	Development RX
0	b210_1, rtl_1, rtl_3, rtl_6, rtl_9	b200_mini_1, pluto_1, rtl_2, rtl_5, rtl_7
1	b200_mini_1, pluto_1, rtl_2, rtl_5, rtl_7	b210_1, rtl_1, rtl_3, rtl_6, rtl_9

Historical confirmation uses four receivers: b200_1 and n210_1, together with rtl_4 and rtl_8. The present protocol does not open six other receivers: b200_2, b200_mini_2, b210_2, n210_2, n210_3, and pluto_2. They were used by the related OOB study and must not be described as new, never-exposed blind receivers.

For each receiver, packet selection uses NumPy's default random generator with seed 20261012, selects 200 rows without replacement per transmitter, and sorts the selected original row indices. For source training, the first 160 selected rows per transmitter train the model and the remaining 40 select its checkpoint. Each evaluation receiver supplies all 200 selected packets per transmitter. Within a fitted model there is no source/evaluation receiver overlap. Across the two complementary folds, a receiver can be a source for one model and an evaluation receiver for another.

The four historical-confirmation receivers have been used in preceding project stages. Earlier signal-level diagnostics were also performed on some of them before the current frozen protocol. Stage-specific sealing prevents a stage's confirmation outcome from changing that stage's fitted checkpoint; it does not erase cumulative development exposure. The public paper should not call historical confirmation an independent blind replication.

2 Training and Architecture Details

The generator seeds for order and added noise are derived from a stable SHA256 seed of the protocol, fold, and seed label. The same derived stream is used for paired arms. With 8,000 source packets,

Item	Fixed value
Source budget per fold	8,000 train + 2,000 validation packets
Fitted instances per configuration	Two folds × three seeds (201, 202, 203)
Batch / epochs	64 / 200, without early stopping
Optimizer	AdamW, weight decay 5×10^{-4}
Learning rate	Peak 10^{-3}; 5-epoch warm-up; cosine decay
Training noise	Per-packet added-noise ratio uniform in [0, 50) dB
Checkpoint	Minimum source RX-equal validation cross-entropy
Reported validation batch	128
Recorded environment	PyTorch 2.5.0+cu124, RTX 4060 Laptop GPU

the implementation runs 125 full batches per epoch; no training packets are dropped. The complete fixed recipe is included in `data/code_13-mixer-cfo-mechanisms__train.py` and the archived baseline configuration.

All strictly in-band views are reconstructed after RMS normalization and FFT masking. FFT magnitude is transformed by $\log(1 + |X|)$ and standardized per window. Amplitude uses $\log(1 + \sqrt{I^2 + Q^2 + 10^{-6}})$; phase is $\text{atan2}(Q, I)/\pi$. The channels are I, Q, normalized FFT magnitude, log amplitude, and phase. Their shared array index combines temporal and frequency coordinates. It is not a physical time alignment between a frequency bin and a time sample.

HStem stage	Input channels	Output channels	Kernel	Stride	Following operation
Shared front-end 1	5	32	7	1	BN + LeakyReLU
Shared front-end 2	32	32	5	1	BN + LeakyReLU
Tokenizer 1	32	48	8	4	BN + GELU
Tokenizer 2	48	64	8	4	BN + GELU
Tokenizer 3	64	64	8	4	BN + GELU
Tokenizer 4	64	64	4	4	None

The first three tokenizer convolutions have padding 2; the last has padding 0. The shared convolutions preserve length with padding 3 and 2. All variants retain 32 tokens, dimension 64, stride 256, and first-token center 127.5 in feature coordinates. Full Patch and Patch-Lite CNN support is 266; HStem and Overlap-Patch support is 350. Global FFT and band projection preclude interpreting these as end-to-end physical RF receptive fields.

Patch-Lite uses groups 8 and intermediate width 80. A data-free search minimizes maximum full-model parameter-count mismatch across both mixer choices, with a 1% tolerance, and breaks ties by fewer groups and intermediate channels. It is a matched-budget control, not an exact capacity or arithmetic-cost match.

3 Cross-Mixer Controls and Receiver Results

The HStem minus Patch-Lite development difference is 2.687 pp for MHA and 1.733 pp for RF-HSTU. Their reported conditional RX 95% intervals are [2.375, 3.083] and [1.400, 2.150] pp, respectively; both have six of six positive fitted pairs. The corresponding historical-confirmation differences are 3.142 and 2.806 pp. The historical-confirmation columns are descriptive because their receivers have prior exposure.

Table 1: Same-stage S15 controls. These RF-HSTU numbers are not the different historical S06 best result.

Mixer	Tokenizer	Parameters	Dev (%)	Hist. (%)
MHA	Full Patch	571,434	95.222	93.685
MHA	Patch-Lite	134,266	94.648	93.367
MHA	HStem	133,690	97.335	96.508
RF-HSTU	Full Patch	580,136	95.888	94.317
RF-HSTU	Patch-Lite	142,968	95.727	93.642
RF-HSTU	HStem	142,392	97.460	96.448

Table 2: Accuracy by receiver, averaged over the applicable fitted repeats (%). Dev receivers are held out from their evaluated fit; historical confirmation is previously exposed.

Role	RX	Patch-Lite	Smooth	Overlap	NoMidBN	Affine	HStem
Dev	b200_mini_1	93.117	95.150	95.200	95.550	96.267	97.200
Dev	b210_1	95.367	96.633	97.383	96.883	97.083	97.933
Dev	pluto_1	90.917	91.083	91.450	94.100	94.383	93.667
Dev	rtl_1	94.917	95.667	96.417	95.933	96.167	97.617
Dev	rtl_2	94.867	95.350	96.783	96.167	96.667	97.533
Dev	rtl_3	93.800	95.233	96.117	94.917	95.000	96.967
Dev	rtl_5	95.500	95.900	97.117	96.567	96.883	97.733
Dev	rtl_6	95.367	96.450	97.350	96.283	96.533	97.500
Dev	rtl_7	96.700	96.900	98.133	98.133	97.967	98.650
Dev	rtl_9	95.933	96.900	97.500	96.917	97.217	98.550
Hist.	b200_1	86.117	86.425	87.142	90.600	89.867	90.867
Hist.	n210_1	96.058	96.817	97.633	97.367	97.092	98.467
Hist.	rtl_4	95.208	96.100	97.125	96.625	96.975	98.208
Hist.	rtl_8	96.083	96.858	97.867	97.242	97.475	98.492

4 Mechanism Contrasts and Local Risk

The four planned development contrasts are reproduced below with the exact stability decision. The interval is not the only criterion. Positive pairs count fold/seed repetitions; the independent clustering unit is RX.

Contrast	Mean pp	RX 95% interval	Positive pairs	Worst RX pp	Signal
Overlap – Smooth	+0.818	[+0.562,+1.053]	6/6	+0.050	Pass
Smooth – Patch-Lite	+0.878	[+0.547,+1.252]	5/6	+0.167	Pass
HStem – AffineOnly	+0.918	[+0.492,+1.295]	4/6	-0.717	Does not pass
AffineOnly – NoMidBN	+0.272	[+0.135,+0.410]	5/6	-0.167	Does not pass

The practical signal requires at least +0.5 pp, both folds positive, at least five of six positive fitted pairs, interval lower bound above zero, and worst-RX difference at least -2 pp. AffineOnly minus NoMidBN falls within the predefined small-effect range on development receivers; it is not evidence of literal equality. On historical-confirmation receivers, that contrast is -0.106 pp with interval [-0.504,+0.292] pp. No multiplicity adjustment is applied; all four planned contrasts are retained.

The file `tx_rx_deltas.csv` contains every recorded fold/seed/TX/RX error-count and error-rate change for these contrasts. Its sign convention is candidate minus reference error: positive values indicate additional errors, even when the corresponding mean accuracy contrast is positive. Local risk must therefore be read with the error sign convention, not inferred from an accuracy ranking. The per-RX table and raw CSV files are provided rather than suppressing cells that do not favor the main model.

5 Parameterization, Conversion, and Precision

The trained Smooth-RoundTrip kernel expands 256 to 340 positions using linear interpolation with `align_corners=False`, scales by $\sqrt{256/340}$, and interpolates back to 256 positions. The resulting map has rank 256. Its singular values span approximately 0.389185 to 0.917756; the eigenvalues of TT^{T} span 0.151465 to 0.842277. Its Fourier-basis off-diagonal energy fraction is approximately 5.309%, so it is not a strictly diagonal frequency rescaling.

A rejected center-crop proposal had rank 194 and would leave 62 base taps inactive. That audit preceded training of the corrected control. No scientifically trained center-crop result is substituted for the full-rank version.

Conversion	Models	Evaluations	Maximum logit error	Pass count
Smooth-RoundTrip to ordinary convolution	6	12,000	0	6/6
HStem BN fold, default TF32 enabled	6	12,000	0.019568	0/6
HStem BN fold, TF32 disabled	6	12,000	5.39×10^{-5}	6/6

The BN tolerance is atol = rtol = 10^{-4}. Predicted classes are unchanged in all listed conversion comparisons. Strict-precision BN folding is a follow-up numerical diagnostic after the original failures, not a replacement record. Folding reduces the independent learned parameter count from 133,690 to 133,338 by absorbing affine terms into convolution weights.

Default-setting interpolation materialization timings have source-model median single-packet p50 values approximately 2.331 ms before and 2.186 ms after conversion, and batch-128 values 14.610 and 14.456 ms. These are separate from the primary tokenizer cost benchmark. Default-setting conversion timings must not be used to claim speed under the strict-precision BN-folding condition. Full per-model/per-repeat measurements and the two precision settings are supplied as CSV files.

6 Noise, Alignment, and Missing Acquisition Information

Frozen source-validation noise results are listed for completeness. Added-noise ratios are relative to observed packet power, not a recovered physical SNR. These tests do not add a new independent receiver evaluation.

Condition	HStem accuracy (%)	Patch-Lite accuracy (%)
Clean	98.175	95.925
Added AWGN 20 dB	86.500	81.008
Added AWGN 10 dB	58.233	52.650
Added AWGN 0 dB	30.583	27.975

HStem's circular -1/+1-sample accuracies are 81.167%/75.933%. Circular -4/+4-sample accuracies are 21.858%/17.775%; the corresponding zero-filled results are 21.867%/17.383%. Patch-Lite also collapses. Circular shifts preserve discrete sample content; energy and FFT-magnitude differences are

numerical-scale in the recorded checks. Locking the magnitude view causes at most a 0.0083 pp accuracy change and does not recover performance.

The existing source HDF5 schema provides fixed IQ windows with label, CFO, and SNR fields. It does not provide neighboring continuous samples, packet-detection timestamps, ground-truth synchronization offsets, or a validated deployment-error distribution. These absences prevent turning the synthetic shift sensitivity into an established synchronization-error mechanism. No synchronization-robust network is proposed or trained to rescue the stress-test outcome.

7 Historical External-Method Context

The following archived same-corpus comparisons use differing implementations, input treatments, recipe variants, and selection procedures. They are retained for context rather than pooled with S22's paired mechanism samples. The full CSV also includes historical internal architectures. Results under an original full-input recipe must not be reinterpreted as strictly in-band performance.

Table 3: Historical same-corpus external-method results. Complete recorded recipes are retained; these are not newly paired tokenizer fits. Accuracy is RX-equal (%).

Recorded configuration	Dev	Hist.	Parameters
FAN_full_official	73.620	75.498	310,481
FAN_full_unified	82.788	82.850	310,481
FAN_ib_unified	89.515	86.529	310,481
IB-RFF_a0.004_ib_unified	94.943	92.208	310,481
IB-RFF_a0.04_full_official	73.552	76.300	310,481
WaveMLP_scaled_full_official	56.682	55.700	594,442
WaveMLP_scaled_full_unified	81.017	78.669	594,442

The contrastive receiver/channel method cited in the main paper is discussed as related work, not included as a newly trained baseline. No result from its different public-data protocol is copied into a numerical ranking. Consequently, this study makes no exhaustive state-of-the-art claim.

8 Overlap With the Related OOB Manuscript

Dimension	Related OOB study	Present contribution boundary
Question	Frozen-model dependence on OOB pathways	Strictly in-band token formation and parameter efficiency
Signal front-end	RMS/TrueIB and five channels	Inherited unchanged; not claimed as new
Local front-end	Kernel-7/kernel-5 CNN and token interface	Inherited unchanged
Tokenizer	Large stride-256 patch projection	Hierarchical replacement; matched controls and interventions
Mixer	OOB fusion and HSTU-style processing	No OOB fusion; ordinary MHA main model; inherited gated mixer as a control

Dimension	Related OOB study	Present contribution boundary
Corpus	Same 20-SDR, 10-TX LoRa source	No new corpus or receiver qualification
Protocol	Different development budgets/selection history	Source-only checkpointing, disclosed reuse; no cross-paper causal accuracy subtraction
Former blind RX	Six receivers evaluated by the prior study	Not opened here; cannot be relabeled never-exposed
Mechanism	Digital pathway dependence, not physical hardware isolation	Parameterization and numerical function checks, not hardware invariance

The supplied related manuscript is titled *Diagnosing Receiver-Sensitive Out-of-Band Dependencies in LoRa Radio-Frequency Fingerprint Identification*, by Ziyang Wang, Chengcheng Han, Jianhuan Chen, and Xin Zhang. The review draft cites the author-supplied manuscript without inventing its publication status or DOI. Its final publication metadata and relationship disclosure must be confirmed by the authors before submission. Its full PDF is not redistributed in this source bundle.

9 Artifact Contents and Audit Boundaries

The source bundle contains the main and supplementary LaTeX files, bibliography, figures, complete frozen aggregate/per-receiver/local-risk tables, inherited implementation snapshots relevant to the stated method, and a SHA256 provenance manifest. Figure and table preparation reads only already-derived CSV/JSON and code files. It performs zero IQ-array reads, checkpoint deserializations, gradient updates, or additional receiver scoring.

The script `prepare_evidence.py` binds source copies to SHA256 hashes and verifies that both role-level mean accuracies recompute from the per-receiver rows. It is a manuscript-data consistency check, not an independent rerun of every historical experiment. Existing original data, weights, licenses, registration files, and exposure history remain authoritative for those experiments.